

Homotopy and Contractibility

Rudra Mukherjee

March 2026

1 Introduction

In this section we will be solving the following problems:

1. Show that the closed interval $[0, 1]$ and the real line $\mathbb{R}$ are contractible and that a contractible space is path connected.
2. A set A is star convex if there is a point a such that the line segment connecting a to b is contained in A for every b in A . Find a star convex set that is not convex; show that star convex sets are simply connected; show that any two paths in a star convex set which have the same initial and final points are actually path homotopic.
3. Let $p : E \rightarrow B$ be a covering map; let B be connected. Show that if $p^{-1}(b_0)$ has k elements for some $b_0 \in B$, then $p^{-1}(b)$ has k elements for every $b \in B$. In such a case, E is called a k -fold covering of B .
4. There is no retraction $r : B_2 \rightarrow S^1$, where B_2 is the unit ball in $\mathbb{R}^2$ and S^1 is the circle.

2 Problem 1

Proof. First, we will show that the closed interval $[0, 1]$ is contractible. Thus we need to construct a homotopy that sends any path in $[0, 1]$ to a constant. Let $\gamma(x)$ be a path that is in $[0, 1]$. Consider the function $H(x, t) = \gamma(x)(1 - t)$. Since $\gamma(x)$ is a path, by definition it is connected, and since $1 - t$ is a polynomial, it is continuous. Thus $H(x, t)$ is continuous with respect to each of its variables. Finally, $H(x, 0) = \gamma(x)(1 - 0) = \gamma(x)$ and $H(x, 1) = \gamma(x)(1 - 1) = 0$. Thus, the identity map is nullhomotopic and therefore $[0, 1]$ is contractible.

We now show that $\mathbb{R}$ is contractible. Letting $\beta(x)$ represent a path that is in $\mathbb{R}$, it follows from what we showed earlier that $H(x, t) = \beta(x)(1 - t)$ is a homotopy from $\mathbb{R}$ to 0. Thus $\mathbb{R}$ is contractible.

Now we show that a contractible space is path-connected. Let C be a contractible space. Consider any two points a and b in C . Since C is contractible, there exists homotopies $H(x, t)$ and $G(x, t)$ such that $H(a, 0) = a$, $H(a, 1) = p$, $G(b, 0) = b$, and $G(b, 1) = p$. Thus if we take the product of $H(x, t)$ and $G(x, 1-t)$, the result will be a homotopy $F(x, t)$ that goes from a to b . By definition, $F(x, t)$ is continuous, so the homotopy is a path from a to b and thus C is path-connected. $\square$

3 Problem 2

Proof. We start with a star convex set that is not convex. Consider the set $E = \{(x, y) | -1 \leq x \leq 1, y = 0\} \cup \{(x, y) | x = 1, 0 \leq y \leq 1\}$ that is a subset of $\mathbb{R}^2$. By definition, the point $(1, 0)$ is a point such that for every point e in E , there exists a line segment from $(1, 0)$ to e . However, a straight line from $(1, 1)$ and $(-1, 0)$ goes through the point $(0, \frac{1}{2})$, which is not in E . Therefore, E is a star convex set that is not convex.

Now we show that every star convex set is simply connected. First, we will show that a star convex set is path connected and then that it is contractible. Suppose that F is a star convex set. Then there exists a point p such that every point f in F has a line segment that connects f to p . Thus, to go from a point a to a point b in F , travel along the line segment from a to p , and then the line segment from p to b . As this is a composition of line segments, it is a continuous path and thus F is path-connected. Now we show that F is contractible. For any path in F , we construct the homotopy H such that for every point c on the path, we travel along the line segment that connects c to p . Thus, the identity map is nullhomotopic and therefore F is simply connected.

Finally, we show that any two paths in a star convex set F that have the same initial and final points are path homotopic. Since F is star convex there is a point e such that every point in F has a line segment that connects to e . Consider any two paths $\alpha(x)$ and $\beta(x)$ in F such that $\alpha(0) = \beta(0) = \alpha$ and $\alpha(1) = \beta(1) = \beta$. To construct the path homotopy, for each point along $\alpha(x)$, we traverse through the line that connects to e , and then backwards from e to a point in $\beta(x)$, so the homotopy is $H(x, t) = \alpha(x)(1 - t) + \beta(x)(t)$. Thus $H(x, 0) = \alpha(x)$, $H(x, 1) = \beta(x)$, $H(0, t) = \alpha$, $H(1, t) = \beta$, and therefore any two paths that have the same initial and final point are path homotopic. $\square$

4 Problem 3

Proof. Consider the set $A = \{b \in B : p^{-1}(b) \text{ has } k \text{ elements}\}$. Note that $A \neq \emptyset$ since b_0 is in A . We will show that A is open and closed in B (this will force A to be B itself).

We first show that A is open. Since E is a covering map, for every point b in B

there exists a neighborhood U that is evenly covered by p . Thus for each point a_0 in A , we have k slices that are homeomorphic to U . Since A is the union of these slices it follows that A is an arbitrary union of open sets and is therefore open.

Now we show that $B \setminus A$ is open. Consider a point b_0 in B that is not in A . Since E is a covering space, the neighborhood U that contains b_0 cannot have k slices that are homeomorphic to U (otherwise it would be in A) and so again we have that $B \setminus A$ is an arbitrary union of open sets, and is thus open.

Since A is open and $B \setminus A$ is open, it follows that A must be equal to B (since B is connected) and therefore if $p^{-1}(b_0)$ has k elements, then $p^{-1}(b)$ has k elements for every $b \in B$. $\square$

5 Problem 4

Proof. Suppose for the sake of contradiction that there is a retraction between B_2 and S^1 . Then such a retraction would induce a surjective homomorphism between the fundamental groups of B_2 and S^1 , but since the fundamental group of B_2 is trivial and the fundamental group of S^1 is isomorphic to $\mathbb{Z}$, such a homomorphism cannot exist. $\square$

References

- [1] Munkres, J. R. (1997). Topology: A first course. Singapore: Prentice Hall International.